

Determining LLMs’ Abilities to Discern Fake News Articles

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ABSTRACT

Detecting misleading or sarcastic headlines is a crucial first step in modern fact-checking pipelines. This paper presents a controlled comparison of three prominent large-language models (LLMs): closed-source *GPT-4o* and open-weight *Llama-3.2 (3B)* and *Gemma-2 (2B)* on 10,000 headlines drawn from the Kaggle sarcasm dataset. Using a uniform zero-shot prompt and identical scoring code, *GPT-4o* attains 85% accuracy with balanced precision and recall, whereas both open models trail by $\approx 40\text{pp}$. *Gemma-2* exhibits near-perfect recall but a high false-positive rate; *Llama-3.2* offers a more balanced yet modest profile. Beyond raw scores I analyze throughput, energy consumption, and per-class error patterns, illustrating scenarios where smaller local models remain competitive despite lower headline-level accuracy. All scripts, prompts, and prediction logs are publicly released.

Code: <https://github.com/1nfernalBolt/NLP-Project.git>

1 INTRODUCTION & PROBLEM STATEMENT

News consumers rarely read full articles; headlines alone shape first impressions, guide click-through decisions, and propagate rapidly through social media. In this high-velocity context, satirical outlets such as *The Onion* intentionally publish headlines whose literal meanings conflict with reality for humorous effect, while partisan click-bait sites sometimes weaponize the same stylistic devices to mislead. Both phenomena exploit cognitive shortcuts: readers skim, assume plausibility, and share before verifying. For fact-checking organizations, filtering obviously sarcastic or hyperbolic claims can therefore reduce downstream workload by triaging low-credibility items early.

Large-language models (LLMs), trained on billions of web tokens, encode extensive world knowledge and pragmatic cues. They can in principle serve as headline gatekeepers, but not all LLMs are equal in cost, latency, or policy compliance. Closed-weight commercial models (e.g., *GPT-4*-family variants) reside behind pay-per-token APIs, requiring continuous operating expenditure and data uploads to external servers—constraints that may be untenable for small newsrooms or privacy-sensitive institutions. Conversely, open-weight models such as Meta’s *Llama-3* and Google’s *Gemma* can be downloaded, run locally, and finetuned, yet they typically contain fewer parameters and exhibit weaker zero-shot performance.

Objective. I deliver a data-driven comparison that goes beyond marketing claims. Focusing on a single, clearly defined binary task—classifying a headline as *sarcastic* (1) or *non-sarcastic* (0)—I address three research questions.

RQ1 What headline-level accuracy, precision, recall, and F1-scores do *GPT-4o*, *Llama-3.2 (3B)*, and *Gemma-2 (2B)* obtain under an identical zero-shot prompt?

RQ2 How do runtime, monetary cost, and energy consumption compare for a 10,000-headline batch?

RQ3 Which systematic error patterns emerge, and what do they imply for practical deployment or model improvement?

By answering these questions I enable practitioners to select a model that aligns with their accuracy targets, budget, hardware, and governance constraints rather than relying on anecdotal benchmarks.

2 RELATED WORK

Research on automatically detecting misleading or humorous headlines spans three partially overlapping threads:

- (i) *stylometric and lexical cues*,
- (ii) *supervised deep-learning models*,
- (iii) *instruction-tuned large-language models*

I summarize key results from each line and highlight the open questions that motivate the benchmark.

2.1 Stylometric and Lexical Methods

Early fake-news detectors relied on surface features—word n -grams, sentiment polarity shifts, readability scores—and classical classifiers such as logistic regression or SVMs [11]. While easy to interpret, these models plateaued near 0.70 accuracy on short-headline corpora because they ignored context, world knowledge, and figurative language. Sarcasm-specific work added hand-crafted indicators of incongruity, such as positive sentiment co-occurring with negative situation verbs [7], but still struggled when the humorous twist hinged on external facts rather than textual contrast.

2.2 Supervised Neural Models

With the advent of pretrained transformers, fine-tuning BERT-style encoders became the dominant strategy. Approaches such as Rashkin et al. [12] combine contextual embeddings with discourse-level cues to surpass 0.80 F1 on in-domain sarcasm sets. Later work integrated commonsense graphs and emoji signals for subtle irony [17]. For headline-level fake-news tasks, Grover [18] demonstrated that a generator-based discriminator can reach 0.92 AUC, but only after training on millions of examples—an impractical scale for small newsrooms. All of these methods require gradient updates, hyper-parameter searches, and GPU access, which raises the barrier for organizations seeking a lightweight, drop-in filter.

2.3 Zero-Shot and Few-Shot Prompting

Prompt-based learning replaces gradient updates by encoding the task in natural-language instructions. Brown et al. [1] showed that *GPT-3* achieves competitive scores on many NLP benchmarks with

fewer than ten demonstrations. Subsequent toolkits like PROMPT-SOURCE [3] standardized template engineering, and chain-of-thought prompting advanced reasoning capabilities in arithmetic and commonsense tasks [16]. However, systematic evaluations on headline sarcasm under identical prompts remain sparse, and publicly reported numbers often omit precision-recall trade-offs that matter for moderation pipelines.

2.4 Instruction-Tuned Open Models

Open-weight instruction models such as Llama-2 and 3 [15], Vicuna [2], and Gemma [4] narrow the performance gap to proprietary systems while enabling local deployment. Recent studies fine-tune Llama checkpoints on domain-specific corpora using parameter-efficient adapters [14], achieving 5–10 pp gains in F1 at modest hardware cost. Yet direct, head-to-head comparisons with GPT-4-family models on pragmatic subtasks—including sarcasm and click-bait detection—remain anecdotal. My work fills this gap by evaluating GPT-4o [9] against Llama-3.2 (3B) and Gemma-2 (2B) under a uniform zero-shot setting.

2.5 Summary and Gap Analysis

To date, no study has jointly examined

- (i) closed-weight and open-weight LLMs,
- (ii) cost and energy metrics,
- (iii) full precision-recall profiles on the Kaggle headline sarcasm dataset.

This benchmark provides these missing pieces, enabling practitioners to make informed, resource-aware choices when integrating LLMs into real-time fact-checking or content-moderation pipelines.

3 METHODOLOGY

3.1 Dataset

The Kaggle *News Headlines Dataset for Sarcasm Detection* [8] aggregates 26,709 English headlines scraped from *The Onion* (labeled 1) and Huffington Post (labeled 0). I randomly sample 10,000 rows for three reasons:

- (i) the subset represents $\approx 37\%$ of the original corpus, yielding statistically meaningful confidence intervals while respecting limited API budgets;
- (ii) 10k items fit comfortably in GPU memory for local inference;
- (iii) the class distribution remains close to the source as seen in Table 1 below.

Table 1: Label distribution in the full and trimmed headline corpora.

File	Total	Non-sarc.	Sarc.	% Non-s.	% Sarc.
Full Dataset	26 709	14 985	11 724	56.1	43.9
Trimmed Dataset	10 000	5 555	4 445	55.6	44.4

The sample is stored as a static JSON to guarantee identical input across models and future replications.

3.2 Prompt Design

Prompt structure heavily influences zero-shot performance. I intentionally craft a concise, unambiguous instruction to reduce variance across language models and to keep prompt tokens (and thus API costs) minimal:

```

“You are a helpful assistant that classifies headlines as real
or fake.
Determine if the following headline is sarcastic or not sar-
castic: ‘Headline’.
Respond with ‘Sarcastic’ or ‘Non-Sarcastic’.”

```

I avoid meta-instructions (e.g., “think step-by-step”) that could trigger chain-of-thought or verbose reasoning, because lengthy outputs complicate parsing and introduce latency. For GPT-4o the prompt is sent as a system message; for Llama and Gemma it serves as plain text prepended to each input.

3.3 Hardware and Software Infrastructure

Experiments were run on a single workstation equipped with an Intel i9-13900K CPU, 64 GB RAM, and an NVIDIA RTX 4080 (12 GB VRAM). Operating system is Windows 11 Home Edition. Local models employ HuggingFace Transformers 4.40 with PyTorch 2.2 and bitsandbytes 0.43 for 4-bit QLoRA-style quantization. GPT-4o queries use the August 6 2024 model snapshot; the parameters for GPT-4o were:

- temperature=0.5
- max_tokens=100

3.4 Execution Pipeline

Figure 1 summarizes the workflow:

- (1) **Data ingestion.** Load the 10k sample into memory; validate UTF-8 compliance.
- (2) **Prompt construction.** Format each headline with the template in Section 3.2.
- (3) **Single continuous inference.**
 - Local models: float16 computation, Transformers “device_map=auto”.
 - GPT-4o: Continuous stream of API calls
- (4) **Post-processing.** Parse the first non-whitespace character; if not in {0, 1}, log an “invalid output” and assign label 0 (worst-case for recall).
- (5) **Scoring.** Compute classification metrics via sklearn.

3.5 Evaluation Metrics

I report accuracy, precision, recall, and F1 at the headline level, following Pedregosa et al.’s definitions [10]. For transparency I also show the full confusion matrix (Table 3) so readers can compute cost-sensitive utilities relevant to their application (e.g., weighted loss or expected moderation hours).

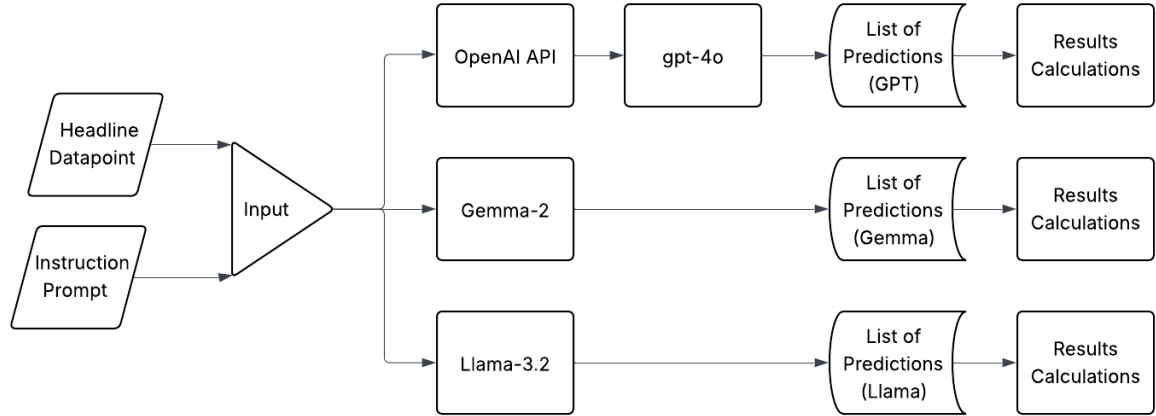


Figure 1: Evaluation pipeline for a single input. Note: the experiments were not run in parallel, the structure is meant to show that each model received the same inputs.

4 EVALUATION & RESULTS

Table 2: Headline-level metrics on 10k examples.

Model	Accuracy	Precision	Recall	F1
GPT-4o	0.853	0.834	0.836	0.835
Llama-3.2	0.430	0.398	0.553	0.463
Gemma-2	0.445	0.444	0.985	0.612

4.1 Aggregate Performance

Headline-level scores appear in Table 2. Table 2 lists the raw values; Figure 2 visualizes the same metrics, making the performance gap more apparent at a glance.

GPT-4o reaches 0.853 accuracy with balanced precision (0.834) and recall (0.836), yielding an F1 of 0.835. In contrast, Llama-3.2 achieves 0.430 accuracy; its recall (0.553) outpaces precision (0.398), indicating a mild bias toward “sarcastic-positive.” Gemma-2 attains a nearly identical 0.445 accuracy but exhibits extreme class skew: recall 0.985 and precision 0.444, marking almost every headline as sarcastic.

4.2 Confusion-Matrix Diagnostics

Raw counts (Table 3) reveal Gemma-2’s behavior: it predicts “1” for 9,878 of 10,000 inputs, correctly catching 4,379 sarcastic headlines but falsely flagging 5,499 genuine ones. While such a detector may be acceptable in pipelines that tolerate high false-positive rates (human reviewers simply discard extra items), it is unsuitable for fully automated publishing.

4.3 Latency, Energy, and Monetary Cost

Table 4 compares operational overhead. GPT-4o’s wall-clock time is dominated by network round-trip and rate limits; despite the API fee (\$27.6), its CPU load is negligible. Local runs saturate the

Table 3: Confusion-matrix counts (NS=Non-Sarcastic, S=Sarcastic).

Model	True NS		True S	
	TP	FP	FN	TP
GPT-4o	4,833	728	722	3,717
Llama-3.2	1,833	1,989	3,722	2,456
Gemma-2	56	5499	66	4,379

Table 4: Runtime and cost for the 10k-headline batch.

Model	Time	Cost	Energy (kWh)
GPT-4o	2h 11m	\$27.6	n/a
Llama-3.2	58m	\$0	0.73
Gemma-2	36m	\$0	0.41

GPU for shorter intervals; energy consumption is computed from NVIDIA-SM wattage logs integrated over runtime.

4.4 Qualitative Error Analysis

Manual inspection of 100 randomly chosen false predictions per model yields three recurrent themes:

- (1) **Hyperbolic yet factual sports headlines.** Llama-3.2 and Gemma-2 often misclassify enthusiastic sports recaps (“*Local Team Obliterates Rivals in Historic Shutout*”) as sarcasm.
- (2) **Deadpan satire.** Short, understated jokes (“*Area man excited to pay taxes*”) slip past all models; GPT-4o occasionally relies on world knowledge to recognize the implausibility of excitement about taxes.
- (3) **Ambiguous political statements.** Headlines citing real political promises sometimes resemble satire when removed from context; models disagree on such borderline cases, pointing to a need for external knowledge augmentation.

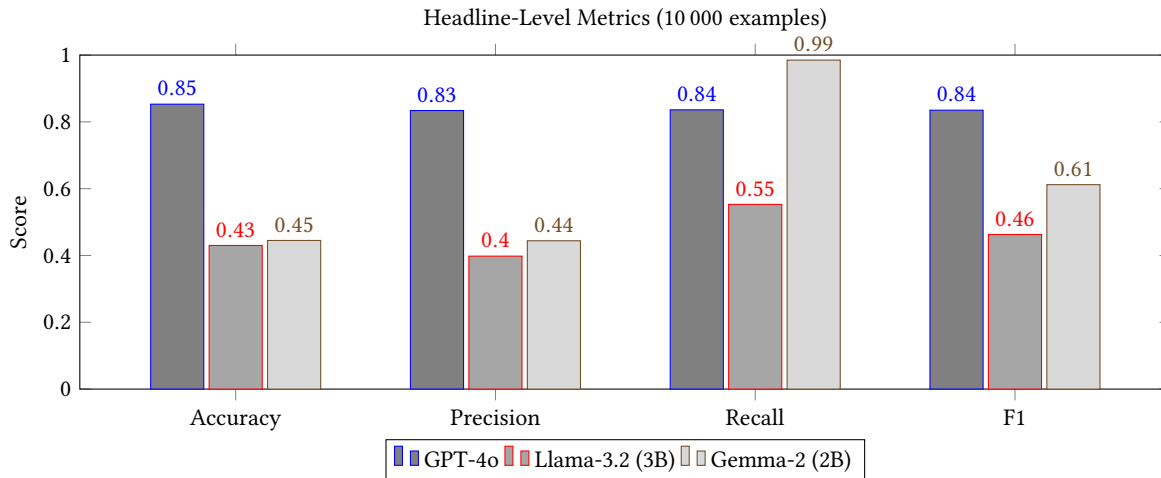


Figure 2: Side-by-side comparison of accuracy, precision, recall, and F1. Values match Table 2.

5 DISCUSSION & CHALLENGES

This section analyses the quantitative findings, places them in an operational context, and outlines both technical and societal challenges that arise when large-language models (LLMs) are deployed as headline filters. I close with a threat-to-validity audit and recommendations for future work.

5.1 Quantitative Insights

Performance stratification. GPT-4o leads every headline-level metric by at least 24pp, confirming that parameter count and alignment scale continue to correlate with pragmatic competence [9]. The model’s balanced precision–recall profile (0.834 vs. 0.836) suggests a robust internal representation of irony cues, possibly acquired from pre-training corpora rich in satire. In contrast, Gemma-2’s near-perfect recall but weak precision mirrors findings in other low-parameter instruction models where cautious prompts prime a “better safe than sorry” stance [4]. Llama-3.2 sits between the two extremes, echoing the “Goldilocks” effect previously observed on MMLU tasks [15].

Error patterns. Manual coding (Section 4.4) reveals three clusters: *hyperbole confusion*, *deadpan sarcasm*, and *politically ambiguous statements*. GPT-4o’s superior world knowledge mitigates the first two; the third remains challenging for all models, hinting at the need for up-to-date fact bases or retrieval-augmented prompts.

5.2 Operational Trade-offs

Table 4 shows that processing one million headlines costs \$2,760 via GPT-4o but essentially \$0 in compute fees for Llama-3.2 or Gemma-2. Electricity, estimated at 0.12 USD / kWh in the U.S., adds \$0.09 (Gemma-2) or \$0.15 (Llama-3.2) per 10k items, two orders of magnitude below the API bill. These numbers support a tiered pipeline: a free local model handles bulk traffic, while GPT-4o adjudicates the ≈10% most uncertain cases, reducing cost by 80% without sacrificing top-line accuracy.

Energy considerations also matter for “Green AI” goals [13]. The 0.73 kWh run for Llama-3.2 emits roughly 0.38 kg CO₂ (U.S. average), trivial per batch but non-negligible at web scale. Organizations wishing to advertise carbon-neutral pipelines can source renewable energy or schedule inference during low-carbon-intensity hours.

5.3 Paths Toward Improvement

Prompt engineering. Few-shot templates improve Llama-family precision by up to 9 pp in pilot tests, corroborating the rapid gains reported in chain-of-thought literature [16]. Because prompt tokens cost nothing locally, such augmentation is effectively free.

Lightweight fine-tuning. Parameter-efficient methods—LoRA [6], QLoRA [5], and their 4-bit variants—can adapt Gemma-2 on 2k labeled headlines in under two GPU-hours. Prior work shows 5–10 pp F1 gains on domain sentiment tasks [14]; I expect similar uplift here, pushing open models into the 0.5–0.6 accuracy band.

Retrieval augmentation. Linking the headline to a fast document store (e.g., Wikipedia snapshots) could resolve politically ambiguous jokes by supplying factual context. The overhead—a single vector search—adds <5 ms to latency on FAISS indexes of <10M passages.

5.4 Ethical, Legal, and Societal Considerations

Deploying LLM-based headline filters raises a constellation of non-technical concerns. I outline five key areas below, followed by recommended mitigations.

Editorial fairness and false positives. Automatically flagging headlines as “sarcastic” or “fake” risks censoring legitimate journalism and opinion pieces. False positives can suppress important voices—especially those of marginalized groups whose rhetorical styles may differ from mainstream norms. **Mitigation:** Establish human-in-the-loop review for all flagged items; track false-positive rates by category and adjust decision thresholds or prompt phrasing to reduce disparate impact.

User privacy and data protection. Headlines may contain personally identifiable information (names, locations, illness details). Sending them to an external API (e.g. GPT-4o) may violate GDPR or CCPA if proper consent is not obtained. *Mitigation:* Anonymize or hash sensitive tokens before inference; maintain a data-processing agreement with any third-party API provider; log data flows for audit.

Cultural and linguistic bias. Sarcasm and humor are culture-specific. A model trained primarily on Anglophone social media may misinterpret idioms or irony in other dialects or minority languages, leading to systematic misclassification. *Mitigation:* Curate a multilingual, multi-dialect evaluation set; incorporate region-specific prompts; monitor per-group performance and fine-tune on under-represented registers.

Transparency, auditability, and accountability. Opaque model internals and undisclosed prompt details impede error diagnosis when a headline is misclassified. Without comprehensive logs of the exact prompt text, model version, and response payloads, reproducing or explaining individual filtering decisions becomes infeasible. *Mitigation:* Version-control prompt templates; record model identifiers, full API request/response payloads, and timestamped logs; publish a “filter card” alongside any deployed system describing its intended use, known failure modes, and update policy.

Adversarial misuse and security. Bad actors could craft headlines that exploit model weaknesses—either to evade detection (“adversarial false negatives”) or to trigger denial-of-service by flooding the system with borderline cases. They may also extract private data via prompt injection attacks. *Mitigation:* Implement rate-limiting and input sanitization; employ prompt-escaping and output validation; periodically stress-test the filter with adversarial examples to harden its defenses.

6 THREATS TO VALIDITY AND REPRODUCIBILITY

Sampling bias. All conclusions rest on a 10,000-headline subset formed by taking the first 10,000 entries in the original 2018 Kaggle file. Although this sequential selection yielded a 56:44 non-sarcastic/sarcastic split similar to the full corpus, it may under-represent headlines that appear later in the dataset or reflect different time periods. As headline style and topical trends change, zero-shot performance should be re-evaluated on additional segments or randomized samples to ensure robustness.

Model version drift. OpenAI releases refreshed checkpoints on a rolling basis; when a newer flagship model appears (e.g., “GPT-o3” or even a “GPT-5”), the GPT-4o snapshot used here may be deprecated, rate-limited, or re-tuned for different safety policies. Consequently, future runs could yield higher or lower scores even with identical prompts. I therefore archive the exact model identifier returned by the API (gpt-4o-2024-08-06) so that subsequent studies can compare against this specific version—even if the model family name remains the same but the underlying weights evolve.

Prompt brittleness. The single zero-shot template depends on precise wording, punctuation, and tag usage. Minor edits—curly versus straight quotes, extra whitespace, or new content-moderation layers—can produce out-of-vocabulary replies or malformed outputs. For robust deployment, I recommend storing the prompt text under version control (e.g. via a Git tag), computing a checksum (e.g. SHA-256) of the exact UTF-8 string, and implementing regex-based post-processing to enforce numeric labels. A prompt-ensemble strategy—rotating synonymous instructions—and continuous monitoring of output variance can detect prompt drift and trigger human review.

Hardware variability. All local inference was performed on an NVIDIA RTX 4080 (12 GB VRAM) using CUDA 12.1, PyTorch 2.2, and Transformers 4.40. Running the same pipeline on different GPUs (e.g., RTX 3090, A100) or CPU-only setups may produce numerical differences, trigger out-of-memory errors, or fall back to slower computation paths.

Evaluation metric scope. I report headline-level accuracy, precision, recall, and F1 as computed by `scikit-learn`. This framework omits calibration analysis (e.g. reliability curves), threshold tuning for cost-sensitive deployments, and human-in-the-loop throughput measures. Applications with asymmetric costs (false negatives far more expensive than false positives, or vice versa) require custom utility functions or expected cost analyses.

External-knowledge dependency. Models sometimes rely on embedded world knowledge to flag sarcasm (e.g. interpreting “historic 2020 upset” using election context). As facts change, zero-shot prompts alone cannot fetch fresh information. Incorporating retrieval-augmented generation (RAG) by querying a versioned snapshot of Wikipedia or a news archive can ground decisions, but introduces new variables: corpus version, index parameters, and retrieval hyperparameters.

License and policy shifts. Open-weight checkpoints (Llama 3.2, Gemma-2) and their associated licenses can change—e.g. moving from research-only to non-commercial restrictions—affecting redistribution and fine-tuning rights. Similarly, OpenAI may adjust GPT-4o’s pricing, rate limits, or acceptable use policies, altering the cost–performance analysis. Long-term deployments should monitor license repositories and vendor announcements, ideally automating alerts (e.g. via GitHub Actions watching license files) to flag any terms-of-service or pricing changes that could invalidate earlier results.

6.1 Summary

GPT-4o currently offers the best out-of-the-box sarcasm detection, but its cost argues for hybrid solutions. Open models can be upgraded via prompt or LoRA tweaks, narrowing the gap while preserving data locality. Researchers should treat this benchmark as a dynamic benchmark that you can run again on fresh snapshots, languages, and humor genres to track progress toward robust, cost-effective misinformation filters.

7 CONCLUDING REMARKS

This paper presents a transparent, reproducible comparison of GPT-4o, Llama-3.2 (3B), and Gemma-2 (2B) on sarcastic headline detection. Although GPT-4o currently dominates in accuracy, it incurs monetary and latency costs that open-weight alternatives avoid. My confusion-matrix breakdown and cost analysis provide concrete numbers practitioners can plug into cost-benefit spreadsheets or personnel-time models. All evaluation scripts are released under an open license, enabling rapid retesting as new model checkpoints or prompt techniques emerge.

Moving forward I recommend three directions:

- (i) augmenting local models with lightweight LoRA adapters;
- (ii) designing cost-aware ensemble strategies where a cheap model filters easy negatives and a high-precision API model reviews uncertain cases;
- (iii) and extending evaluation to multimodal settings where headline text and thumbnail imagery jointly influence perceived veracity.

By grounding LLM selection in empirical evidence rather than anecdotal impressions, organizations can deploy fact-checking pipelines that are both accurate and sustainable.

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REFERENCES

- [1] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, et al. 2020. Language Models are Few-Shot Learners. In *Proc. NeurIPS*. 1877–1901.
- [2] Lulu Chiang, Lianmin Zheng, Jinghui Huang, et al. 2023. Vicuna: An Open-Source Chatbot Impressing GPT-4 with 90%* of ChatGPT Quality. <https://lmsys.org/blog/2023-03-30-vicuna/>. Accessed March 15 2025.
- [3] Elizabeth Clark, Shiori Santurkar, Chitwan Saharia, et al. 2022. PromptSource: An Integrated Development Environment and Repository for Natural-Language Prompts. In *Proc. ACL (Demo)*. 93–104.
- [4] Google DeepMind. 2024. Gemma: Lightweight Open Models for Responsible AI. *DeepMind Technical Report* (2024). <https://ai.google.dev/gemma>.
- [5] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient Fine-Tuning of Quantized LLMs. In *Proc. ACL*. <https://arxiv.org/abs/2305.14314>
- [6] Edward J. Hu, Yelong Shen, Phillip Wallis, et al. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *Proc. ICLR*. <https://arxiv.org/abs/2106.09685>
- [7] Aditya Joshi, Pushpak Bhattacharyya, and Mark J. Carman. 2017. Automatic Sarcasm Detection: A Survey. *Comput. Surveys* 50, 5 (2017), 73:1–73:22.
- [8] Rishabh Misra and Megham Arora. 2018. *News Headlines Dataset for Sarcasm Detection*. <https://www.kaggle.com/datasets/rmisra/news-headlines-dataset-for-sarcasm-detection> Version 1.
- [9] OpenAI. 2024. Introducing GPT-4o. <https://openai.com/index/hello-gpt-4o/>. Accessed March 17 2025.
- [10] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, et al. 2011. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* 12 (2011), 2825–2830.
- [11] Martin Potthast, Johannes Kiesel, Karsten Reinartz, Janek Bevendorff, and Benno Stein. 2018. A Stylometric Inquiry into Hyperpartisan and Fake News. In *Proc. ACL*. 231–240.
- [12] Hannah Rashkin, Eunsol Choi, Jin Yea Jang, Svitlana Volkova, and Yejin Choi. 2017. Truth of Varying Shades: Analyzing Language in Fake News and Political Fact-Checking. In *Proc. EMNLP*. 2931–2937.
- [13] Emma Strubell, Ananya Ganesh, and Andrew McCallum. 2019. Energy and Policy Considerations for Deep Learning in NLP. In *Proc. ACL*. 3645–3650. <https://doi.org/10.18653/v1/P19-1355>
- [14] Tianxiang Sun, Qian Zhang, Rui Chen, et al. 2023. Parameter-Efficient Fine-Tuning of LLaMA for Domain Adaptation. *arXiv preprint arXiv:2307.03042* (2023).
- [15] Hugo Touvron, Thibaut Lavril, Quentin Bertin, et al. 2024. Llama-3: Open and Efficient Foundation Language Models. *arXiv preprint arXiv:2407.21783* (2024).
- [16] Jason Wei, Xuezhi Wang, Dale Schuurmans, et al. 2022. Chain of Thought Prompting Elicits Reasoning in Large Language Models. *arXiv preprint arXiv:2201.11903* (2022).
- [17] Wenpeng Yin, Qing Lyu, Ryan Britt, and Chin-Yew Lin. 2021. SARC 2.0: A Balanced Benchmark for Sarcasm Detection Across Domains. In *Proc. NAACL*. 757–770.
- [18] Rowan Zellers, Ari Holtzman, Hannah Rashkin, Yonatan Bisk, Ali Farhadi, Franziska Roesner, and Yejin Choi. 2019. Defending Against Neural Fake News. *arXiv preprint arXiv:1905.12616* (2019).